

PepperCone Shadow Box Material Notes

Fabric vs. paint for blocking ambient light around the cone

Main idea

- The shadow box needs to block ambient light and background detail around the rig so the reflected image reads as higher contrast.
- Two material options: paint the enclosure matte black, or line it with black fabric.
- Start with cheap black fabric for the first test. It is reported to absorb light better than ordinary paint and is far easier to iterate with during testing.
- Only move to a specialty ultra-black product if standard fabric does not give enough contrast.

Research notes

Standard black paint has real limits

- Even good flat/matte black paint still has some surface sheen and reflects more light than people expect, especially at a grazing viewing angle.

Fabric generally outperforms paint for this use

- Deep-pile fabrics and purpose-built blackout cloth trap light in their fiber structure instead of reflecting it directly, and are reported to outperform standard black paint for killing reflections.
- A side-by-side optical comparison of black materials found that black velvet, felt, and specialized fabrics like Musou Black fabric absorb more visible light and scatter it more evenly across viewing angles than typical black paints, including chalkboard and spray-on acrylic black.
- Fabric is also more practical for an iterative build: it can be draped, clamped, or Velcro'd around the rig and swapped out between tests, instead of repainting the enclosure each time we change the design.

Specialty ultra-black products

- Musou Black fabric and flock sheet products absorb up to roughly 99.9% of visible light, making them the most effective publicly available option, but at a higher cost than basic felt or blackout cloth.
- Vantablack, the most absorptive material referenced in the literature (about 99.965% absorption), is not available for general/consumer use and is not a realistic option for this project.

Plan

- Step 1: Build the first shadow box test using cheap black felt or blackout curtain fabric draped

around the sides and back of the rig.

- Step 2: Fire up the cone and compare contrast against the current no-enclosure baseline.
- Step 3: If contrast is still weak, test a denser fabric in the same setup.
- Step 4: Only consider a specialty ultra-black product if standard fabric options are not enough, given the added cost and harder application.